THREAT INTELLIGENCE

MALICIOUS PCAP INVESTIGATION

A practical investigator walkthrough

LAB FORMAT Network traffic analysis • Malware intelligence • Evidence correlation

This exercise places participants in the role of a digital investigator. They must reconstruct a suspicious download from packet-capture evidence, identify the downloaded artifact, and enrich the finding using malware-intelligence data.

Evidence provided: A PCAP containing network communication associated with the delivery of a malicious archive.

Expected approach: Preserve the evidence, establish the download chain, extract the artifact in a controlled environment, and correlate static-analysis results with the network evidence.

1. OBJECTIVE

The objective of this lab is to develop a repeatable threat-intelligence investigation workflow using network traffic as the initial source of evidence. Participants will analyse the supplied PCAP to identify the infrastructure that hosted a suspicious ZIP archive, reconstruct the download, inspect the archive contents, and use a malware-intelligence service to profile the extracted file.

By the end of the exercise, participants should be able to:

- Identify suspicious domains, IP addresses, HTTP requests, and server metadata in a packet capture.
- Determine the name of a downloaded archive and identify its contents.
- Extract a transferred object from network traffic while maintaining evidence integrity.
- Use VirusTotal to review detection classification, first-seen information, contacted URLs, and dropped-file observations.
- Correlate network indicators with file-based intelligence and document defensible findings.

2. BENEFICIARIES

This practical is intended for SOC analysts, threat-intelligence analysts, incident responders, digital-forensics investigators, malware analysts, and cybersecurity trainees. It is particularly useful for participants learning to move from raw network telemetry to evidence-backed threat intelligence.

3. INVESTIGATION SCENARIO

An endpoint within an organisation communicated with an external web server and downloaded a ZIP archive. The security team has recovered a PCAP covering the relevant communication, but the incident details are unknown. You are assigned as the investigator. Your task is to examine the traffic, establish which domain and server delivered the archive, identify the downloaded file and its contents, and enrich the extracted artifact using VirusTotal. You must support each conclusion with evidence from the PCAP or the intelligence report rather than relying on assumptions.

4. LAB REQUIREMENTS

- Wireshark or an equivalent packet-analysis tool.
- A controlled analysis VM with archive inspection utilities.
- Access to VirusTotal for file-hash lookup or submission, subject to organisational policy.
- A working directory for extracted evidence, notes, hashes, and screenshots.

5. INVESTIGATION WORKFLOW

1. Preserve and scope the evidence

Work on a copy of the supplied PCAP. Record its filename, acquisition context, and SHA-256 hash. Note the capture time range and avoid modifying the original evidence.

2. Identify the suspicious download

Open the PCAP in Wireshark. Review DNS queries and HTTP conversations, then look for responses containing archive content or a ZIP filename. Use stream reconstruction to establish the request and response sequence.

3. Attribute the hosting infrastructure

Record the requested hostname, destination IP, relevant HTTP response headers, and any server-identification strings. Distinguish the web-server product from the server software/version string returned in the headers.

4. Recover and inspect the archive

Use Wireshark's object-export functionality where supported, or reconstruct the HTTP stream and save the response body. Verify that the recovered object is a ZIP archive, calculate its SHA-256 hash, and list its contents without executing anything.

5. Enrich the extracted file

Submit the file to VirusTotal only if permitted by your organisation's data-handling policy. Where possible, search by SHA-256 first. Review the detection verdict, first-seen timestamp, contacted URLs, and dropped files.

6. Correlate and report

Compare the PCAP indicators with the VirusTotal report. Record the source of each finding, preserve screenshots or exported records, and clearly distinguish observed facts from analytical conclusions.

Operational safety note

Do not execute the recovered file on a production endpoint. Analyse it in an isolated, disposable environment. VirusTotal submissions may be shared with security partners and may expose submitted files or metadata. Do not upload confidential, proprietary, personal, or otherwise restricted samples unless authorised. If submission is not permitted, search using the file hash or use an approved internal sandbox.

6. INVESTIGATION QUESTIONS

Answer the following questions using evidence from the packet capture and the VirusTotal intelligence report. Include the relevant packet, HTTP header, extracted-file, or VirusTotal evidence for each response.

Q1. Which internet domain served as the source of the suspicious ZIP archive?

Q2. What filename identifies the ZIP archive retrieved by the endpoint?

Q3. After inspecting the archive, what is the exact filename of the document contained inside it?

Q4. Which web-server software is identified in the response from the external host?

Q5. What server software/version string is disclosed in the HTTP response headers?

Q6. After safely extracting the document, use VirusTotal to search for its hash or submit the sample if authorised. What malware classification is reported?

Q7. According to VirusTotal, what is the earliest recorded observation date and time for this file?

Q8. How many distinct URLs are listed as contacted by the analysed file?

Q9. How many files are reported as dropped by the analysed file?

7. EVIDENCE AND REPORTING CHECKLIST

- PCAP filename and SHA-256 hash recorded.
- Suspicious domain and destination IP documented.
- Relevant HTTP request/response or stream identified.
- Archive recovered and its SHA-256 hash calculated.
- Archive contents listed; no file executed.
- Web-server product and disclosed version string recorded separately.
- VirusTotal result captured with the lookup hash and report date.
- All answers linked to supporting evidence.

8. EXPECTED LEARNING OUTCOME

Participants should be able to produce a concise incident-intelligence record that links a suspicious network download to its hosting domain, server characteristics, recovered archive, embedded document, and malware-intelligence profile. The emphasis is on evidence preservation, repeatable analysis, and defensible attribution—not merely obtaining the answers.

9. INSTRUCTOR ANSWER KEY

The values below are the expected answers supplied for this exercise. Keep this section separate from the participant handout if the lab is being assessed.

Investigation item	Expected result
Q1 — Hosting domain	**attirenepal.com**
Q2 — Downloaded archive	**documents.zip**
Q3 — Archive member	**chart-1530076591.xls**
Q4 — Web-server software	**LiteSpeed**
Q5 — Server version string	**PHP/7.2.34**
Q6 — Malware classification	**Trojan**
Q7 — First-seen timestamp	**2021-09-25 03:26:03**
Q8 — Contacted URLs	**4**
Q9 — Dropped files	**9**

Instructor validation note: Confirm that the VirusTotal results correspond to the exact extracted file or its SHA-256 hash. The first-seen time, URL count, and dropped-file count can vary between report versions or intelligence-provider views; the supplied values are the lab's expected reference values.